

Food Donation & NGO Connect Platform

A relational-database network that connects surplus food to nearby NGOs

REVIEW 1

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Connecting surplus food to NGOs

Every day, restaurants, households and event organisers discard edible food while NGOs struggle to find nearby donors in time to collect it.

This project proposes a web platform backed by a relational database. Donors list surplus food and NGOs post their requirements; SQL queries match them by location, quantity and expiry window, and database triggers alert NGOs automatically.

Every pickup is tracked with an audit trail, giving a verified donor-NGO network and less urban food wastage.



Discover

NGOs and volunteers see surplus food in real time.



Match

SQL queries pair listings with nearby NGOs.



Track

Every pickup is logged with a full audit trail.

02

Food rescue is slow and untracked

1

No shared record

Donor listings and NGO needs sit in phone contacts and chat groups, with no common database.

2

Slow, manual coordination

Volunteers phone around to match food with an NGO, which does not scale across a city.

3

Food spoils while waiting

Cooked meals expire quickly, so delays in finding a nearby NGO mean food is discarded.

4

No traceability or trust

Nobody can verify donors or NGOs, or confirm that a pickup was completed and delivered.

Manual coordination, no structure

CURRENT PRACTICE

Phone calls and WhatsApp groups

Basic donation-listing apps

Volunteers matching by contacts

NGOs checking for new posts

Informal pickups

LIMITATION

Messages get buried and there is no searchable record

Donors can post, but nothing matches or ranks listings

Depends on who is available and who remembers what

No automatic alert when new surplus food appears

No delivery tracking, audit trail or verification

One network for every donation



Central listings

Donors post type, quantity, location and expiry in one structured database.



SQL matching engine

Joins and proximity filters rank NGOs by distance, quantity, expiry and priority score.



Automated triggers

Triggers update listing status and alert NGOs when a matching donation is added.



Verified network

Admins verify donors and NGOs before they can list or claim food.



Pickup lifecycle

Status moves Listed → Matched → Picked up → Delivered, with volunteers assigned.



Audit trail

Every match logs the donor, NGO and criteria behind the assignment.

OBJECTIVES

05

What we set out to achieve

01

Build one central relational record of donors, NGOs, volunteers and listings

02

Match donations to NGOs with SQL by location, quantity and expiry

03

Notify NGOs automatically using database triggers

04

Track every pickup from listing to delivery with an audit log

05

Verify donors and NGOs to build a trusted network

06

Give admins tools to monitor listings, resolve disputes and view reports

Scope of the project

IN SCOPE

- ✓ Web app for donors, NGOs and volunteers
- ✓ Structured listings: food type, quantity, location, expiry
- ✓ Normalised (3NF) schema with keys, constraints and indexes
- ✓ SQL matching on proximity, quantity and expiry window
- ✓ Triggers and stored procedures for status updates and alerts
- ✓ Pickup assignment and delivery status tracking
- ✓ Audit table logging every match
- ✓ Admin panel for verification and monitoring

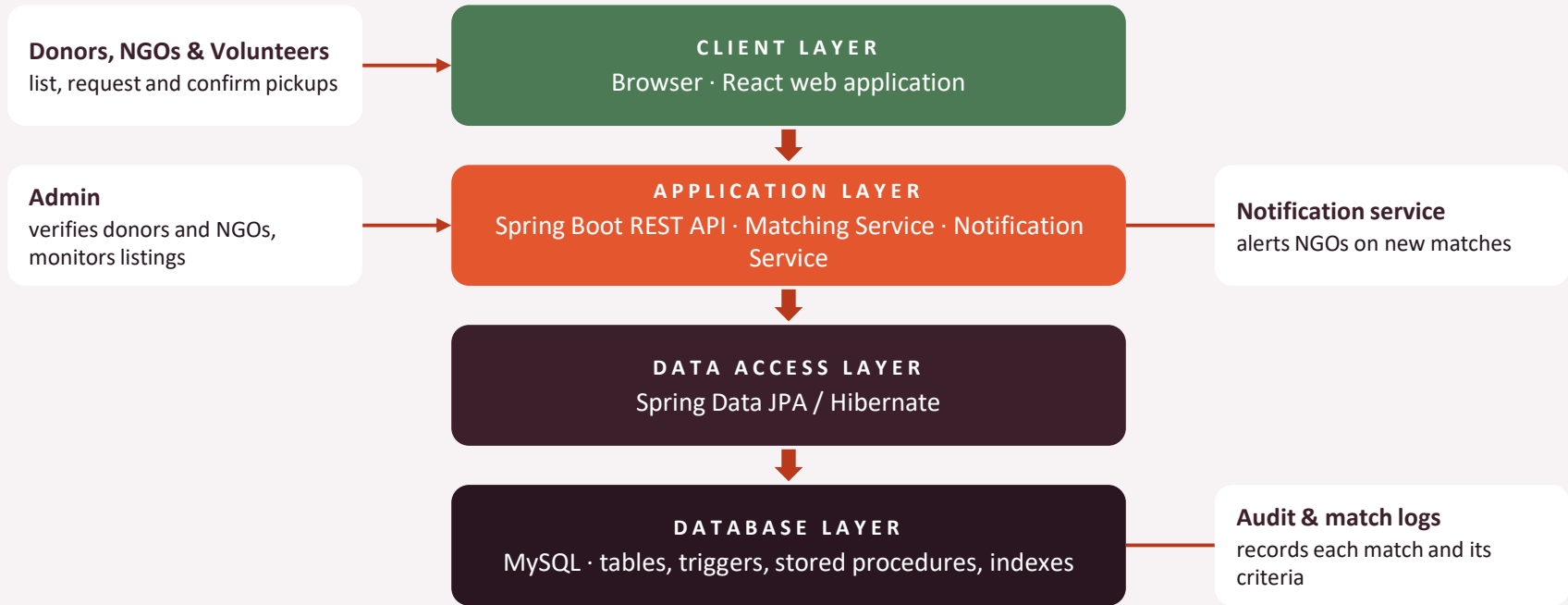
OUT OF SCOPE (PHASE 1)

- Native Android / iOS apps
- Courier or transport service
- Payments or monetary donations
- Demand forecasting and multi-city rollout (future scope)

Technology stack

FRONTEND**React (web application)****AUTOMATION****AFTER INSERT / UPDATE triggers****BACKEND****Java with Spring Boot (REST APIs)****QUERY TOOLS****MySQL Workbench****DATABASE****MySQL (relational DBMS)****DEV TOOLS****VS Code, IntelliJ IDEA, Postman****DATA ACCESS****Spring Data JPA / Hibernate****VERSION CONTROL****Git & GitHub****MATCHING****SQL joins, stored procedures and
priority scoring****SCALING****AWS RDS / Azure SQL for city-wide
hosting**

System architecture



Thank You

Food Donation & NGO Connect Platform

REVIEW 1 · ABSTRACT, SCOPE, TECH STACK, ARCHITECTURE

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